Direct Current Circuit Analysis

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Official examination scope

PRC Board Resolution No. 40 s. 2024 promulgates the enhanced TOS for the REE subjects; consult its official Annex A for the complete scope, competencies, and weights. This lesson selects foundational DC network analysis for instruction. It does not claim that these examples occur frequently or carry a particular exam weight.

Learning objectives

- Apply Ohm's law to a resistive circuit and retain voltage, current, and resistance units.
- Use series or parallel reduction only when the connection topology supports it.
- Write and check Kirchhoff current and voltage equations with stated reference directions.

1. Start with the circuit model

State that the circuit is a DC lumped resistive model and label reference current directions and polarities. Ohm's law is V = I x R. Use volts, amperes, and ohms consistently. A negative solved current means the actual direction is opposite the reference arrow; it is not automatically an invalid circuit.

2. Reduce simple resistor groups

Resistors in a single series path carry the same current and have equivalent resistance R_total = R1 + R2 + ... Resistors connected across the same two nodes are in parallel and have reciprocal equivalent resistance: 1/R_eq = 1/R1 + 1/R2 + ... Verify the node connections before applying either shortcut.

3. Apply conservation at nodes and loops

Kirchhoff's current law follows charge conservation: total current entering a node equals total current leaving. Kirchhoff's voltage law says the algebraic sum of voltage changes around a closed loop is zero. Choose a direction, record each sign consistently, solve the equations, and substitute the result back into the node and loop balances.

Essential relationships

- Ohm's law: V = I x R
- series resistors: R_eq = sum of resistances
- parallel resistors: 1/R_eq = sum of reciprocal resistances
- junction rule: sum of currents in = sum of currents out
- loop rule: algebraic sum of voltage changes = 0

Series loop with a voltage check

Question: A 20 V source feeds 4 ohms and 6 ohms in series. Find the current and each resistor voltage drop.
1. R_total = 4 + 6 = 10 ohms.
2. I = V/R_total = 20/10 = 2 A.
3. The drops are 2 x 4 = 8 V and 2 x 6 = 12 V.
Answer: Current = 2 A; voltage drops = 8 V and 12 V.
Check: The drops sum to 20 V, matching the source. Source power is 20 x 2 = 40 W; resistor power is 2^2 x 4 + 2^2 x 6 = 40 W.

Current balance at a node

Question: A node receives 5 A. Two outgoing branch currents are 2 A and 1.5 A. Find the third outgoing current.
1. Write incoming current = outgoing current: 5 = 2 + 1.5 + I3.
2. Solve I3 = 5 - 3.5 = 1.5 A.
Answer: The third branch carries 1.5 A away from the node.
Check: The outgoing total is 2 + 1.5 + 1.5 = 5 A.

Common misconceptions

- Adding parallel resistances directly instead of using reciprocal conductances.
- Treating two components as parallel because they look side by side rather than checking their
- nodes.
- Changing sign conventions midway through a loop equation.
- Rejecting a negative current instead of interpreting it relative to the chosen reference
- direction.
- Using DC resistance rules for an AC impedance problem without checking the model.

Practice patterns

- Compute series or parallel equivalent resistance and total current.
- Find a branch current using a node balance or current division.
- Write a loop equation with voltage source polarity and resistor drops.
- Check a proposed answer by current, voltage, unit, or power balance.

Official scope references

Official scope: Enhanced Tables of Specifications for REE — Annex A — Professional Regulatory Board of Electrical Engineering, Professional Regulation Commission — https://www.prc.gov.ph/sites/default/files/2024-40%20Annex%20A.pdf

Instructional references

Instructional reference: University Physics Volume 2, Section 10.3: Kirchhoff's Rules — OpenStax, Rice University — https://openstax.org/books/university-physics-volume-2/pages/10-3-kirchhoffs-rules

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No claim about question frequency or examination weighting is made.